THE ALTERNATING CECH COMPLEX COMPUTES DERIVED GLOBAL SECTIONS OF AN UNBOUNDED COMPLEX OF MODULES UNDER VANISHING HYPOTHESES ON A BASIS OF OPENS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites eight results living in earlier chapters; each is entered in assembly.external_references as a navigation pointer with its canonical tag, and none of them replaces a load-bearing step of this proof (the unbounded limit argument is fully inside the span).

Editorial note. The proof cites no bibliography item inside the counted span, so the card carries no bibliography excerpt.

Allowed standard prerequisites (editorial).

- Comparison of the usual Cech complex with derived global sections for bounded below complexes. Stacks Project, Cohomology of Sheaves, Lemma lemma-cech-complex-complex-computes (tag 0FLH). Supplies the bounded below case once the alternating Cech complex is shown quasi-isomorphic to the usual one.
- Spectral sequences of a double complex and their convergence. Stacks Project, Homology, Lemmas homology-lemma-ss-double-complex (tag 0130) and homology-lemma-first-quadrant-ss (tag 0132). Justify the first-page comparison and the convergence step in the bounded below case.
- Mittag-Leffler criterion for the cohomology of a limit of complexes. Stacks Project, Homology, Lemma homology-lemma-apply-Mittag-Leffler-again (tag 070E); also More on Algebra, Lemma more-algebra-lemma-apply-Mittag-Leffler-again (tag 0918). Lets the eventual constancy of the left-hand cohomology be transferred to the limit on the right-hand side.

Source reference. Original lemma 08C1: sources/stacks-chow/tags.txt, source lines 10586–10586. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01FM: sources/stacks-chow/tags.txt, source lines 1875–1875. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0130: sources/stacks-chow/tags.txt, source lines 1418–1418. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0132: sources/stacks-chow/tags.txt, source lines 1420–1420. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Source reference. Original lemma 0FLH: sources/stacks-chow/tags.txt, source lines 19497–19497. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 071B: sources/stacks-chow/tags.txt, source lines 8985–8985. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 070E: sources/stacks-chow/tags.txt, source lines 8953–8953. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0918: sources/stacks-chow/tags.txt, source lines 11436–11436. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{U} : X = \bigcup_{i \in I} U_i$ be a finite open covering. Let $\mathcal{F}^\bullet$ be a complex of $\mathcal{O}_X$-modules. Let $\mathcal{B}$ be a set of open subsets of X. Assume*

1. *every open in X has a covering whose members are elements of $\mathcal{B}$,*
2. *we have $U_{i_0 \ldots i_p} \in \mathcal{B}$ for all $i_0, \ldots, i_p \in I$,*
3. *for every $U \in \mathcal{B}$ and $p > 0$ we have*
   (a) *$H^p(U, \mathcal{F}^q) = 0$,*
   (b) *$H^p(U, \mathrm{Coker}(\mathcal{F}^{q-1} \to \mathcal{F}^q)) = 0$, and*
   (c) *$H^p(U, H^q(\mathcal{F})) = 0$.*

Then the map

$$Tot(\check{\mathcal{C}}^\bullet_{alt}(\mathcal{U}, \mathcal{F}^\bullet)) \longrightarrow R\Gamma(X, \mathcal{F}^\bullet)$$

of Lemma 08C1 is an isomorphism in $D(Ab)$.

Proof. First assume $\mathcal{F}^\bullet$ is bounded below. In this case the map

$$\mathrm{Tot}(\check{\mathcal{C}}^\bullet_{alt}(\mathcal{U}, \mathcal{F}^\bullet)) \longrightarrow \mathrm{Tot}(\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}^\bullet))$$

is a quasi-isomorphism by Lemma 01FM. Namely, the map of double complexes $\check{\mathcal{C}}^\bullet_{alt}(\mathcal{U}, \mathcal{F}^\bullet) \to \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}^\bullet)$ induces an isomorphism between the first pages of the second spectral sequences associated to these complexes (by Homology, Lemma 0130) and these spectral sequences converge (Homology, Lemma 0132). Thus the conclusion in this case by Lemma 0FLH and assumption (3)(a).

In general, by assumption (3)(c) we may choose a resolution $\mathcal{F}^\bullet \to \mathcal{I}^\bullet = \lim \mathcal{I}^\bullet_n$ as in Lemma 071B. Then the map of the lemma becomes

$$\lim_n \mathrm{Tot}(\check{\mathcal{C}}^\bullet_{alt}(\mathcal{U}, \tau_{\geq -n}\mathcal{F}^\bullet)) \longrightarrow \Gamma(X, \mathcal{I}^\bullet) = \lim_n \Gamma(X, \mathcal{I}^\bullet_n)$$

Here the arrow is in the derived category, but the equality on the right holds on the level of complexes. Note that (3)(b) shows that $\tau_{\geq -n}\mathcal{F}^\bullet$ is a bounded below complex satisfying the hypothesis of the lemma. Thus the case of bounded below complexes shows each of the maps

$$\mathrm{Tot}(\check{\mathcal{C}}^\bullet_{alt}(\mathcal{U}, \tau_{\geq -n}\mathcal{F}^\bullet)) \longrightarrow \Gamma(X, \mathcal{I}^\bullet_n)$$

is a quasi-isomorphism. The cohomologies of the complexes on the left hand side in given degree are eventually constant (as the alternating Čech complex is finite). Hence the same is true on the right hand side. Thus the cohomology of the limit on the right hand side is this constant value by Homology, Lemma 070E (or the stronger More on Algebra, Lemma 0918) and we win. □